TrES-1b

R-0228

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_250916` · created 2026-07-17T03:13:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460934.6906 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.092 +/- 0.052
Transit depth (Rp/Rs)^2	0.85 +/- 0.97 [%]
Orbital Inclination (inc)	85.07 +/- 1.43
Airmass coefficient 1 (a1)	0.842 +/- 0.064
Airmass coefficient 2 (a2)	0.16 +/- 0.26
Scatter in the residuals of the lightcurve fit is	7.91 %
Best Comparison Star	#3 - [385, 120]
Optimal Aperture	3.24
Optimal Annulus	9.75
Transit Duration (day)	0.047 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.092 ± 0.052 vs 0.1358 (deviation 0.8σ)
Duration fit vs published	0.047 d vs 0.1042 d (deviation 1.71σ)
Mid-time O–C	-7.2 min
Depth SNR	1.7
Residual scatter	7.91 %

Light curve

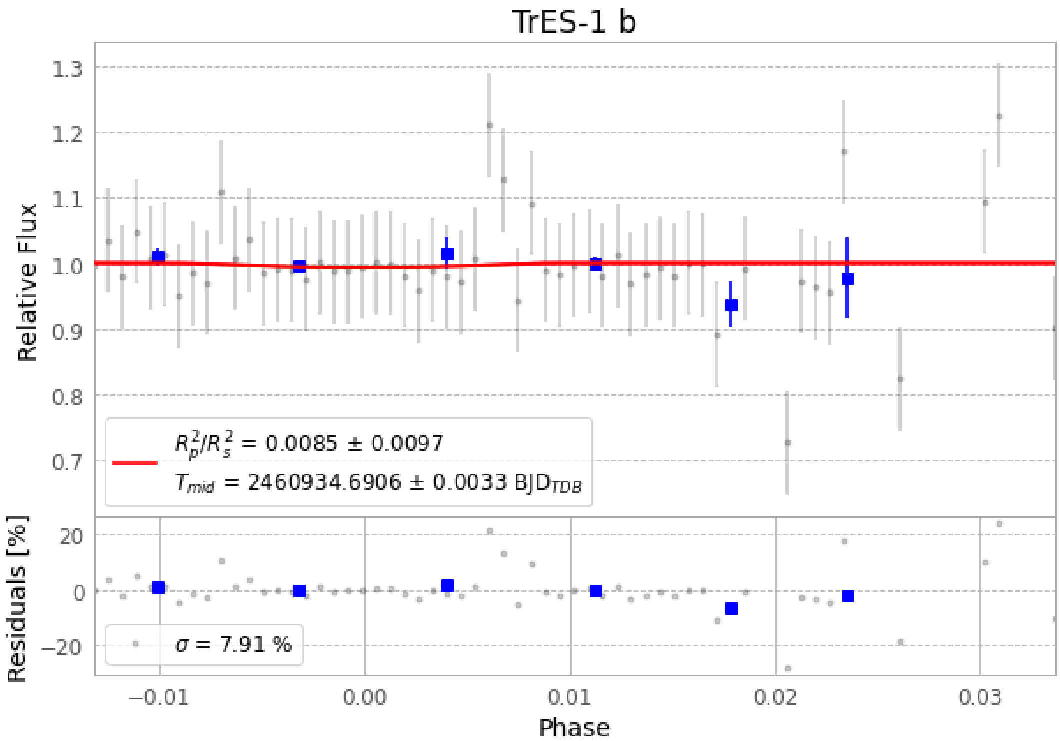

Figure 1 — FinalLightCurve_TrES-1 b_2025-09-15.png (SHA-256 6188b41c6e9ead82…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 255], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a979b7294b82aa3d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

Data provenance

69 FITS frames (45 MB), `TRES-1250916033315.FITS` → `TRES-1250916065715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250916.log (bc6e3f2dcb91…), FinalLightCurve_TrES-1 b_2025-09-15.png (6188b41c6e9e…), FinalLightCurve_TrES-1 b_2025-09-15.csv (27ef60365f64…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-17 03:13Z.